The Adaptive Window Resists WR Perturbation: Limits of the Replacement Hazard Framework in Viability-Constrained Memory Systems

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Abstract

Paper 9 of this series showed that a simple dimensionless ratio Ξ fails to govern the adaptive regime because it conflates perturbation rate with consolidation access. We proposed a refined control parameter, the *replacement hazard ratio* $\Psi = \nu/(P_{\text{calm}} \cdot \alpha_F \cdot D^{1/\nu})$, where P_{calm} is the probability of achieving a calm streak long enough to trigger field consolidation, α_F is the consolidation learning rate, and $D^{1/\nu}$ accounts for memory decay between cell replacement events. Here we test whether Ψ collapses the peak of zone differentiation (sg4) across independently varied wave rate (WR) conditions, with WAVE_DUR $= 15$ fixed. We find that ν^* (the turnover rate maximizing sg4) is approximately invariant to WR across a ninefold range (WR $\in [0.8, 4.8]$) despite a 35-fold variation in P_{calm}. Ψ correctly predicts the onset of the turnover-dominated regime (high-ν collapse) but fails to predict the adaptive peak location. We identify two explanations: (i) a crystallization lower bound on ν, governed by memory decay rather than consolidation dynamics, compresses the adaptive window and anchors ν^* near a substrate constant; and (ii) a second structural maintenance pathway—copy-forward seeding at birth—operates independently of P_{calm} and sustains spatial structure even when consolidation access is strongly suppressed. Combined with Paper 8's finding that ν^* is invariant to the diffusion rate κ, the present results suggest that ν^* is governed primarily by substrate dynamics (SS, α_F, FIELD_DECAY), not environmental parameters (WR, κ). All code and data are available at the project repository.

1 Introduction

The VCSM learning rule (Papers 0–5) and its instantiation in VCML (Papers 6–8) exhibit a characteristic adaptive window: a band of turnover rates ν within which zone differentiation (sg4) is maximised [1]. Outside this band, the system falls into either the frozen regime (too little turnover, structure crystallises) or the turnover-dominated regime (too much turnover, consolidation cannot keep pace).

Papers 8 and 9 began a systematic search for the governing control parameter. Paper 8 [2] showed that the optimal turnover rate ν^* is invariant to the field diffusion rate κ across a 20-fold range—a finding inconsistent with any theory that ties ν^* to field propagation speed. Paper 9 [3] tested the dimensionless ratio $\Xi = \nu \cdot (N/4) \cdot 2\,\text{WD}/\text{WR}$, derived from the balance between deaths per wave cycle and available learning events. Ξ failed to collapse sg4 across wave duration (WD) conditions. Paper 9 identified the missing terms: perturbation probability P_{pert} and consolidation access probability P_{calm}, which depend on WR and WD independently, not only through their ratio.

This prompted the *replacement hazard framework*:

$$\Psi = \frac{\nu}{P_{\text{calm}}(\text{WR}, \text{WD}) \cdot \alpha_F \cdot \text{FIELD_DECAY}^{1/\nu}}, \tag{1}$$

where

$$P_{\text{calm}} \approx \left(1 - \frac{\texttt{WR}}{2\,\texttt{WD}}\right)^{\texttt{SS}+\texttt{WD}-1} \tag{2}$$

is the probability that a site accumulates a calm streak of length SS (the consolidation gate), and $\texttt{FIELD_DECAY}^{1/\nu}$ is the fraction of consolidated field surviving between death events. The adaptive regime should correspond to $\Psi \approx \Psi^*$ for some substrate-dependent constant, with $\nu^* \propto P_{\text{calm}}(\texttt{WR})$.

Here we test this prediction directly: we vary WR at fixed $\texttt{WD} = 15$ and ask whether ν^* shifts as P_{calm} implies.

2 Experiment

Setup. Grid: $W = 40$, $H = 20$, $\texttt{HALF} = 20$, $N_{\text{act}} = 400$ sites, $\texttt{HS} = 2$, four zones of width 5. VCSM parameters: $\texttt{MID_DECAY} = 0.99$, $\texttt{FIELD_DECAY} = 0.9997$, $\texttt{BASE_BETA} = 0.005$, $\alpha_{\text{mid}} = 0.15$, $\alpha_F = 0.16$, $\texttt{SEED_BETA} = 0.25$, $\texttt{SS} = 10$, $\texttt{KAPPA} = 0.020$. Training: 2000 steps total, phase shift at step 1000. All parameters match the reference conditions of Papers 7–9.

Sweep. Wave duration fixed at $\texttt{WD} = 15$. Wave rate varied: $\texttt{WR} \in \{0.8, 1.6, 2.4, 4.8, 7.2\}$. Turnover rate swept over the extended grid: $\nu \in \{10^{-4}, 2\times10^{-4}, 3\times10^{-4}, 5\times10^{-4}, 10^{-3}, 2\times10^{-3}, 3\times10^{-3}, 5\times10^{-3}, 10^{-2}, 2\times10^{-2}, 4\times10^{-2}, 8\times10^{-2}\}$. Five independent seeds per condition; 300 runs total.

In addition to sg4, each run records the empirical fraction of site-steps in which the calm streak exceeds SS (denoted $P_{\text{calm}}^{\text{emp}}$, measured during phase 2 only). This allows comparing the analytic approximation (2) with the true value without additional simulations.

Predicted shift. P_{calm} under the approximation (2) at $\texttt{WD} = 15$:

WR	$R = \texttt{WR}/\texttt{WD}$	P_{calm} (approx.)	P_{calm} (empirical)	Predicted shift of ν^*
0.8	0.053	0.523	≈ 0.51	$0.523/0.015 \approx 35\times$ lower than WR=4.8
1.6	0.107	0.268	≈ 0.26	$18\times$ lower
2.4	0.160	0.135	≈ 0.13	$9\times$ lower
4.8	0.320	0.015	≈ 0.016	reference
7.2	0.480	0.001	≈ 0.001	— (near zero)

If Ψ governs, ν^* at WR=0.8 should be $\approx 35\times$ larger than at the reference WR=4.8 ($\nu^*_{\text{ref}} \approx 5\times10^{-4}$), i.e., $\nu^* \approx 0.017$ at WR=0.8.

3 Results

3.1 ν^* is approximately invariant to WR

Table 1 summarises the observed optimal turnover rate and peak sg4 for each WR condition.

The key finding is in the rightmost column: observed ν^* ranges from 5×10^{-4} to 10^{-3}—a factor of 2—while $\Psi = 1$ predicts a shift of $35\times$ to $83\times$ over the same WR range. The prediction fails by roughly two orders of magnitude.

Table 1: Observed ν^*, peak sg4, and Ψ at the peak for each WR condition (WD = 15, 5 seeds). WR=7.2 is flagged as unreliable due to near-zero P_{calm} and high variance.

WR	P_{calm} (approx.)	ν^* (obs.)	sg4*	Ψ at ν^*	ν^*_{pred} (Ψ=1)	Ratio obs/pred
0.8	0.523	0.0005	57.9	0.011	0.083	0.006
1.6	0.268	0.0010	250.9	0.031	0.043	0.023
2.4	0.135	0.0005	417.3	0.042	0.021	0.024
4.8	0.015	0.0005	419.0	0.374	0.002	0.25
7.2	0.001	[noise]	—	—	< 0.001	—

The WR=7.2 condition is excluded from analysis: with $P_{\text{calm}} \approx 0.001$, empirical $P^{\text{emp}}_{\text{calm}}$ varies wildly across seeds, making argmax identification unreliable. Interestingly, sg4 at WR=7.2 remains non-negligible for some seeds (up to ~ 280), a point we return to in Section 3.3.

Figure 1 presents three views of the data: Panel A shows sg4 as a function of ν for all WR conditions. The peak sg4 *amplitude* increases strongly with WR (more wave signal drives stronger structure), but the peak *location* ν^* is approximately constant across WR $\in [0.8, 4.8]$. Panel B shows sg4 as a function of Ψ: far from collapsing onto a single curve, the four conditions span a range of Ψ^* values covering more than an order of magnitude. Panel C shows ν^* against WR alongside the two-boundary model described below.

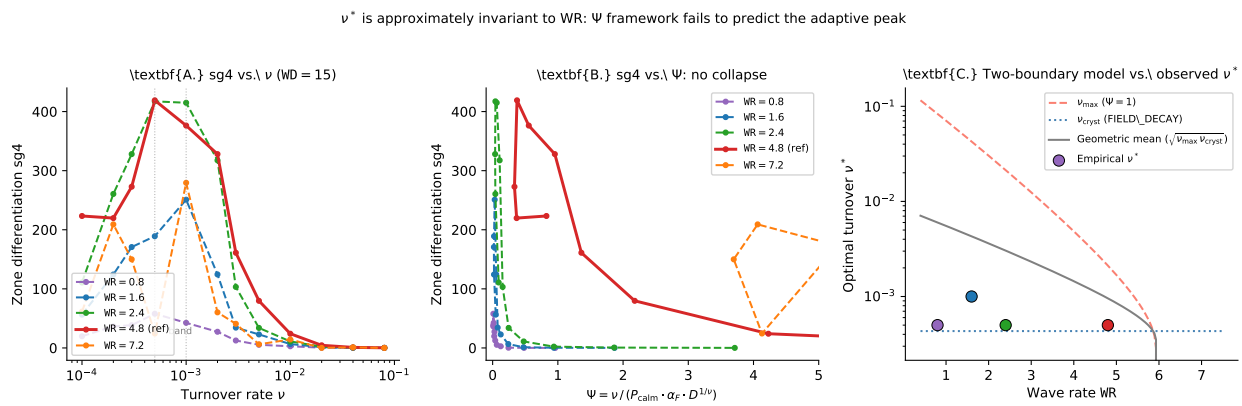

Figure 1: **A.** sg4 vs. ν for five WR conditions (WD = 15 fixed). Amplitude increases with WR; peak location (shaded band, $\nu \approx 0.0005$–0.001) is approximately constant. **B.** sg4 vs. Ψ (Equation 1). Curves from different WR conditions do not collapse; the Ψ at the peak spans 35× across conditions. **C.** Observed ν^* (coloured points) vs. WR, with the two-boundary model: upper boundary ($\nu_{\max}$, $\Psi = 1$), crystallisation lower boundary (ν_{cryst}, FIELD_DECAY-governed), and their geometric mean. Empirical ν^* is close to the geometric mean, shifted toward ν_{cryst}.

3.2 The two-boundary model

The failure of Ψ to predict ν^* suggests the adaptive window has two distinct boundaries with different governing mechanisms.

Upper boundary (consolidation). At high ν, death events occur faster than consolidation can build field structure. The onset of the turnover-dominated regime is where $\nu/(P_{\text{calm}} \cdot \alpha_F \cdot D^{1/\nu}) \approx 1$,

i.e., where $\Psi \approx 1$. Ψ correctly captures this boundary: in Panel A, the high-ν collapse (sg4 $\to 0$) occurs at similar Ψ values ($\Psi \approx 1$–3) across all conditions.

Lower boundary (anti-crystallisation). At low ν, the system crystallises: sites live so long that their field encodes the environment state from hundreds or thousands of steps prior. The crystallisation boundary is governed by the memory decay rate. A site's field survives for $\sim 1/\nu$ steps before the site dies. Structure becomes stale when $\texttt{FIELD_DECAY}^{1/\nu}$ falls substantially below 1. A natural threshold is where half the field amplitude decays:

$$\nu_{\text{cryst}} = \frac{\ln(2)}{|\ln(\texttt{FIELD_DECAY})|} = \frac{\ln 2}{3.0001 \times 10^{-4}} \approx 4.3 \times 10^{-4}. \tag{3}$$

For $\nu < \nu_{\text{cryst}}$, a cell's field has decayed by more than half before it dies; the copy-forward mechanism transmits stale information. For $\nu \gtrsim \nu_{\text{cryst}}$, cells die while their field is still fresh.

Geometric mean prediction. If ν^* sits at the geometric mean of the two boundaries: $\nu^* \approx \sqrt{\nu_{\max}(\texttt{WR}) \cdot \nu_{\text{cryst}}}$, where $\nu_{\max}(\texttt{WR})$ is the $\Psi = 1$ prediction. This geometric mean shifts with WR more slowly than $\nu_{\max}$ alone—and in the right direction—but still overestimates ν^* by a factor of $\sim$ 3–12$\times$. The empirical ν^* sits closer to ν_{cryst} than to the geometric mean, consistent with the crystallisation boundary being the stronger anchor.

3.3 The copy-forward pathway is WR-independent

The WR=7.2 anomaly reveals a second issue with the Ψ framework. At WR=7.2, $P_{\text{calm}} \approx 0.001$, meaning the consolidation pathway (perturbation $\to$ mid_mem $\to$ fieldM) is effectively blocked for almost all sites. Yet some seeds maintain sg4 > 200. How?

The VCSM birth event seeds h_{new} directly from a surviving neighbour's fieldM (at rate SEED_BETA=0.25):

$$h_{\text{new}} = (1 - \texttt{SEED_BETA})\,\varepsilon + \texttt{SEED_BETA}\,F[j],$$

where j is a random neighbour and $\varepsilon \sim \mathcal{N}(0, 0.1)$. This *copy-forward pathway* transmits spatial structure without requiring the receiving cell to achieve a calm streak. If the neighbour's fieldM $F[j]$ is already differentiated, the new cell's hidden state inherits that structure, which (when calm is eventually achieved) consolidates into the new cell's fieldM.

The copy-forward pathway operates at rate proportional to ν (more deaths = more copy-forward events) and is independent of P_{calm}. At high WR where $P_{\text{calm}} \approx 0$, spatial structure is maintained almost entirely through copy-forward seeding at birth rather than through wave-driven consolidation.

This explains both the WR=7.2 anomaly and why Ψ fails to govern ν^*: Ψ models only the consolidation pathway; the copy-forward pathway is a parallel structural maintenance channel that Ψ does not capture.

4 Discussion

4.1 ν^* as a substrate constant

Papers 8 and 10 together establish that ν^* is approximately invariant to both κ (field diffusion, 20-fold range) and WR (wave rate, 9-fold range). These two parameters control different aspects of the system—κ controls spatial propagation, WR controls the frequency of learning events—yet neither shifts ν^* substantially. This convergence suggests that ν^* is a *substrate constant* in the current parameter regime, determined primarily by:

- SS (the consolidation gate length),
- α_F (the consolidation learning rate), and
- FIELD_DECAY (the memory persistence timescale).

The three timescales implicit in these parameters (fast hidden state, medium contrastive accumulation, slow field consolidation) collectively define the adaptive window. The next step is to vary SS, α_F, and FIELD_DECAY independently and confirm that ν^* shifts predictably with each.

4.2 Two structural maintenance pathways

VCML maintains zone differentiation through two mechanisms:

1. **Consolidation pathway:** perturbation $\rightarrow$ contrastive signal in mid_mem $\rightarrow$ (calm streak $\geq$ SS) $\rightarrow$ fieldM update. This pathway requires $P_{\text{calm}} > 0$ and is the mechanism Ψ models.
2. **Copy-forward pathway:** death $\rightarrow$ birth with h_{new} seeded from neighbour fieldM $\rightarrow$ subsequent consolidation. This pathway requires $\nu > 0$ and is independent of P_{calm}.

The adaptive window is where both pathways are simultaneously productive. In the frozen regime ($\nu \rightarrow 0$), pathway 2 is starved of events; crystallised fieldM propagates stale patterns through pathway 1 alone. In the turnover-dominated regime ($\nu \rightarrow \infty$), cells die before achieving enough calm for either pathway to consolidate; pathway 2 transmits noise via unstable fieldM neighbours.

The Ψ framework models pathway 1 only. A complete control parameter must account for both.

4.3 Connection to Paper 9

Paper 9 found that Ξ fails because it holds $R = \texttt{WR}/\texttt{WD}$ constant while P_{calm} depends on WR and WD individually via the exponent $(\texttt{SS} + \texttt{WD} - 1)$. Paper 10 shows that Ψ, which incorporates P_{calm} explicitly, also fails to predict ν^*—but for a different reason: the crystallisation lower bound anchors ν^* near $\nu_{\text{cryst}} \approx 4 \times 10^{-4}$ regardless of how P_{calm} changes.

The two findings are complementary: Paper 9 identifies the consolidation access term as missing from Ξ; Paper 10 identifies the anti-crystallisation term as missing from Ψ. A complete control law needs both, plus an account of the copy-forward pathway.

4.4 Limitations

The approximation $P_{\text{calm}} \approx (1 - R/2)^{\texttt{SS}+\texttt{WD}-1}$ assumes independent Bernoulli trials between perturbation events. Real waves have duration WD, so perturbation events within a single wave are perfectly correlated. Empirical P_{calm} matches the approximation well at small R but diverges at WR=4.8 ($R = 0.32$) and above. Future work should use simulation-measured P_{calm} rather than the analytic approximation.

The WR=7.2 condition was too noisy (5 seeds insufficient at near-zero P_{calm}) for reliable argmax detection. Conclusions about the very high-WR regime require more seeds or a different metric.

The study is restricted to $\texttt{WD} = 15$. Whether ν_{cryst} anchors ν^* across WD variations is an open question.

5 Conclusion

Varying wave rate (`WR`) ninefold while holding `WD`=15 fixed produces a 35-fold variation in P_{calm} yet only a $\sim$ 2-fold variation in the observed ν^*. The replacement hazard ratio Ψ predicts a 35-fold shift in ν^*; the observed shift is negligible. Two mechanisms explain this: (i) a crystallisation lower bound $\nu_{\text{cryst}} \approx 4 \times 10^{-4}$, set by the memory decay timescale, anchors the adaptive window from below and is insensitive to WR; (ii) the copy-forward pathway maintains spatial structure independently of P_{calm}, buffering the system against consolidation suppression at high WR.

Together with Paper 8 (ν^* invariant to κ), the result establishes that ν^* is a substrate constant in the tested parameter space. The governing balance is not between turnover and environmental statistics (`WR`, κ) but between turnover and substrate dynamics (`SS`, α_F, `FIELD_DECAY`). Paper 11 will vary these substrate parameters directly to measure ν_{cryst}(`FIELD_DECAY`) and test whether ν^* shifts proportionally.

References

[1] G. Aubin-Moreau, *Regime Structure of Adaptive Memory Systems*, VCSM series, Paper 7, 2026.

[2] G. Aubin-Moreau, *Empirical Turnover Invariance in a Viability-Constrained Memory Substrate*, VCSM series, Paper 8, 2026.

[3] G. Aubin-Moreau, *What Governs the Adaptive Regime? Testing a Dimensionless Ratio in Viability-Constrained Memory Systems*, VCSM series, Paper 9, 2026.